

## GRANBURY, TX, USA

WMO: 720286

Lat: 32.444N

Lon: 97.817W

Elev: 237

StdP: 98.51

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53977

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -4.9       | -2.6 | -12.2                         | 1.3 | 0.9  | -9.9 | 1.7 | 3.2  | 11.1                  | 11.8 | 9.9 | 10.4 | 2.3                   | 320  | 0.457 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 11.9                   | 39.7            | 22.5 | 38.0 | 22.7 | 37.1 | 22.6 | 25.1                | 32.5 | 24.7 | 32.0 | 24.2 | 31.6 | 3.9                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.3                            | 18.6 | 27.8 | 22.8 | 18.1 | 27.5 | 22.4 | 17.6 | 27.4 | 77.6          | 32.6 | 75.7 | 31.9 | 74.1 | 31.7 | 27.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 9.3           | 8.4  | 7.6 | DB                         | -8.9 | 40.9               | 2.2 | 1.5                                                | -10.6 | 42.0       | -11.9 | 42.8       | -13.1 | 43.7       | -14.8 | 44.7 | (4) |
| (5)               |      |     | WB                         | -9.7 | 25.8               | 2.2 | 0.8                                                | -11.2 | 26.3       | -12.5 | 26.8       | -13.7 | 27.2       | -15.3 | 27.7 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 19.6   | 8.2   | 10.6  | 15.2  | 19.2  | 23.6  | 28.3  | 30.0  | 30.5  | 26.1  | 19.9  | 14.0  | 9.1  | (6)  |
| (7)  |                                                                                  | DBStd       | 9.02   | 5.39  | 6.10  | 5.36  | 4.55  | 3.88  | 2.73  | 2.50  | 2.64  | 3.50  | 4.95  | 5.32  | 5.33 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 283    | 97    | 61    | 16    | 2     | 0     | 0     | 0     | 0     | 0     | 2     | 22    | 83   | (8)  |
| (9)  |                                                                                  | HDD18.3     | 1193   | 315   | 224   | 124   | 43    | 7     | 0     | 0     | 0     | 1     | 41    | 147   | 290  | (9)  |
| (10) |                                                                                  | CDD10.0     | 3789   | 42    | 77    | 178   | 278   | 422   | 548   | 619   | 634   | 483   | 310   | 142   | 54   | (10) |
| (11) |                                                                                  | CDD18.3     | 1658   | 2     | 7     | 28    | 69    | 171   | 298   | 361   | 376   | 233   | 91    | 17    | 3    | (11) |
| (12) |                                                                                  | CDH23.3     | 20702  | 33    | 77    | 259   | 676   | 1780  | 3781  | 4976  | 5282  | 2690  | 959   | 152   | 36   | (12) |
| (13) |                                                                                  | CDH26.7     | 10936  | 4     | 14    | 63    | 222   | 747   | 2049  | 2919  | 3187  | 1339  | 354   | 33    | 5    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.2    | 3.1   | 3.4   | 3.8   | 4.0   | 3.6   | 3.5   | 3.0   | 2.7   | 2.5   | 2.9   | 3.0   | 2.9  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 905    | 56    | 65    | 80    | 80    | 103   | 96    | 49    | 71    | 79    | 91    | 69    | 58   | (15) |
| (16) |                                                                                  | PrecMax     | 1386   | 106   | 205   | 183   | 178   | 268   | 299   | 131   | 199   | 227   | 241   | 220   | 265  | (16) |
| (17) |                                                                                  | PrecMin     | 506    | 15    | 0     | 1     | 10    | 40    | 25    | 0     | 1     | 15    | 14    | 9     | 5    | (17) |
| (18) |                                                                                  | PrecStd     | 217    | 26    | 50    | 45    | 44    | 54    | 79    | 36    | 49    | 55    | 60    | 53    | 54   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 27.0  | 28.5  | 30.8  | 34.8  | 36.3  | 39.7  | 41.2  | 41.3  | 38.0  | 34.6  | 30.1  | 27.2 | (19) |
| (20) |                                                                                  |             | MCWB   | 14.3  | 16.0  | 18.1  | 19.8  | 22.2  | 22.5  | 22.1  | 22.2  | 22.5  | 21.1  | 18.7  | 16.6 | (20) |
| (21) |                                                                                  | 2%          | DB     | 23.3  | 25.9  | 28.5  | 31.3  | 34.2  | 37.6  | 39.2  | 40.0  | 36.2  | 32.3  | 27.3  | 23.6 | (21) |
| (22) |                                                                                  |             | MCWB   | 13.9  | 16.3  | 18.0  | 19.4  | 22.2  | 22.4  | 22.4  | 22.5  | 22.3  | 21.2  | 17.7  | 15.2 | (22) |
| (23) |                                                                                  | 5%          | DB     | 21.0  | 23.0  | 26.3  | 29.2  | 32.6  | 36.3  | 37.8  | 38.5  | 34.8  | 30.4  | 25.0  | 21.2 | (23) |
| (24) |                                                                                  |             | MCWB   | 13.7  | 15.5  | 17.1  | 19.2  | 21.6  | 22.5  | 22.6  | 23.0  | 22.3  | 20.4  | 17.3  | 14.8 | (24) |
| (25) |                                                                                  | 10%         | DB     | 18.0  | 21.0  | 24.1  | 27.3  | 31.1  | 35.0  | 36.6  | 37.3  | 33.1  | 28.5  | 22.6  | 18.2 | (25) |
| (26) |                                                                                  |             | MCWB   | 11.9  | 14.8  | 16.3  | 18.7  | 21.4  | 22.8  | 22.7  | 22.8  | 22.0  | 19.4  | 16.4  | 13.0 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 18.9  | 20.0  | 20.8  | 22.6  | 25.0  | 25.8  | 25.7  | 25.4  | 25.1  | 24.1  | 21.8  | 19.5 | (27) |
| (28) |                                                                                  |             | MCD B  | 21.3  | 25.2  | 26.9  | 28.5  | 32.0  | 33.4  | 33.0  | 33.3  | 32.1  | 29.5  | 26.5  | 23.7 | (28) |
| (29) |                                                                                  | 2%          | WB     | 16.8  | 18.8  | 19.7  | 21.8  | 24.0  | 24.9  | 25.1  | 24.9  | 24.4  | 23.0  | 19.9  | 17.8 | (29) |
| (30) |                                                                                  |             | MCD B  | 20.5  | 23.0  | 25.3  | 27.6  | 30.5  | 32.2  | 32.4  | 32.8  | 30.8  | 28.6  | 23.8  | 21.7 | (30) |
| (31) |                                                                                  | 5%          | WB     | 14.8  | 17.4  | 19.0  | 21.0  | 23.3  | 24.3  | 24.6  | 24.4  | 23.9  | 22.4  | 18.9  | 15.8 | (31) |
| (32) |                                                                                  |             | MCD B  | 19.2  | 21.3  | 24.1  | 26.6  | 29.6  | 31.5  | 32.0  | 32.4  | 30.1  | 27.7  | 22.8  | 19.1 | (32) |
| (33) |                                                                                  | 10%         | WB     | 12.4  | 15.6  | 17.9  | 20.2  | 22.7  | 23.9  | 24.1  | 24.0  | 23.4  | 21.5  | 17.7  | 13.8 | (33) |
| (34) |                                                                                  |             | MCD B  | 17.7  | 19.7  | 22.6  | 25.5  | 28.7  | 31.1  | 31.6  | 32.1  | 29.7  | 26.5  | 21.8  | 17.9 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 13.1  | 12.5  | 12.8  | 12.8  | 11.6  | 11.4  | 11.8  | 11.9  | 11.8  | 13.0  | 12.8  | 12.4 | (35) |
| (36) |                                                                                  |             | MCDBR  | 17.4  | 16.0  | 14.6  | 14.9  | 13.4  | 13.0  | 13.1  | 13.4  | 13.1  | 13.7  | 14.6  | 16.3 | (36) |
| (37) |                                                                                  |             | MCWBR  | 9.7   | 8.9   | 6.8   | 6.7   | 4.5   | 2.9   | 2.7   | 2.7   | 3.4   | 5.4   | 7.4   | 9.5  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 14.3  | 13.5  | 12.1  | 11.3  | 10.8  | 10.7  | 11.0  | 11.4  | 10.7  | 10.9  | 11.3  | 12.5 | (38) |
| (39) |                                                                                  |             | MCWBR  | 9.6   | 8.9   | 6.8   | 6.2   | 4.6   | 3.3   | 3.0   | 2.9   | 3.2   | 5.1   | 7.6   | 9.2  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.309  | 0.319 | 0.350 | 0.388 | 0.421 | 0.428 | 0.435 | 0.428 | 0.399 | 0.359 | 0.334 | 0.314 | (40) |      |
| (41) |                                                                                  | taud        | 2.509  | 2.489 | 2.420 | 2.338 | 2.311 | 2.325 | 2.347 | 2.348 | 2.374 | 2.469 | 2.475 | 2.498 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 915    | 939   | 929   | 902   | 870   | 859   | 852   | 854   | 867   | 882   | 876   | 888   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 86     | 96    | 111   | 126   | 131   | 129   | 125   | 123   | 115   | 97    | 88    | 82    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 3.00   | 3.55  | 4.64  | 5.60  | 5.99  | 6.72  | 6.82  | 6.21  | 5.23  | 4.12  | 3.16  | 2.64  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.30   | 0.56  | 0.35  | 0.47  | 0.59  | 0.60  | 0.46  | 0.41  | 0.42  | 0.62  | 0.41  | 0.34  | (45) |      |

## Historical Trends

|      |                         | DBAvg |     | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|-----|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       |     | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e) | (f)     | (g)    | (h)     | (i)   | (j)   | (k)         | (l)     | (m)     |         |
| (46) | Station Only            | N/A   | N/A | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | (46)    |
| (47) | Regional (45 neighbors) | N/A   | N/A | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | (47)    |

Nomenclature: See separate page